A B C D E F
10 11 12 13 14 15

# Digital Logic Design
## Homework 1
## Covers Chapters 1-2

1. Convert to hexadecimal and then to binary:

   (a) $757.25_{10}$ (b) $123.17_{10}$ (c) $356.89_{10}$ (d) $1063.5_{10}$

a) $1011110101.01_{(2)}$ b) $111011.00101_{(2)}$ c) $101101101.111_{(2)}$ d) $10000100111.1_{(2)}$

2. Convert to base 6: $3BA.25_{14}$ (do all of the arithmetic in decimal).

$3BA.25_{(14)} = 752.17_{(10)} = 3252.1004_{(6)}$

3. Add and multiply in binary:

(a) 1111 and 1010 (b) 110110 and 11101 (c) 100100 and 10110

(a) 10010110 (b) 11000011110 (C) 11000011000

4. Add the following numbers in binary using 2's complement to represent negative numbers. Use a word length of 6 bits (including sign) and indicate if an overflow occurs.

   (a) 21 + 11 (b) (–14) + (–32) (c) (–25) + 18 (d) (–12) + 13 (e) (–11) + (–21)

(a) overflow! (b) overflow! (c) NOT overflow! (d) NOT overflow! (e) NOT overflow!

5. Construct a table for 7-3-2-1 weighted code and write 3659 using this code.

3 6 5 9

0100 0111 0110 1010
or
0011

| Decimal | 7 | 3 | 2 | 1 |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 |
| 2 | 0 | 0 | 1 | 0 |
| 3 | 0 | 1 | 0 | 0 or 0011 |
| 4 | 0 | 1 | 0 | 1 |
| 5 | 0 | 1 | 1 | 0 |
| 6 | 0 | 1 | 1 | 1 |
| 7 | 1 | 0 | 0 | 0 |
| 8 | 1 | 0 | 0 | 1 |
| 9 | 1 | 0 | 1 | 0 |

6. Prove the following theorems algebraically(using boolean algebra laws):

(a) X(X′ + Y) = XY (b) X + XY = X

(c) XY + XY′ = X (d) (A + B)(A + B′) = A

(a) $X(X'+Y) = XY$
$\Rightarrow XX'+XY = 0+XY = XY$

(b) $X+XY = X$
$\Rightarrow X(1+Y) = X\cdot 1 = X$

(c) $XY+XY' = X(Y+Y') = X\cdot 1 = X$

(d) $(A+B)\cdot(A+B') = AA+AB'+BA+BB'$
$= A + A(B'+B) + 0$
$= A+A = A$

7. Simplify each of the following expressions by applying *one* of the (simplification)theorems. State the theorem used.

(a) $X'Y'Z + (X'Y'Z') = X'Y'$ (b) $(AB' + CD)(B'E + CD) = CD + AB'E$

(c) $ACF + AC'F = AF$ (d) $A(C + D'B) + A' = A' + C + D'B$

(e) $(A'B + C + D)(A'B + D) = A'B + D$ (f) $(A + BC) + (DE + F)(A + BC') = A + BC + DE + F$

8. Draw a circuit that uses only one AND gate and one OR gate to realize each of the following functions:

(a) $(A + B + C + D)(A + B + C + E)(A + B + C + F)$

(b) $WXYZ + VXYZ + UXYZ$

7. (a) $X'Y'Z + (X'Y'Z')$
$= X'Y'$

(b) $\underset{Y}{(AB'} + \underset{X}{CD)}\ \underset{Z}{(B'E} + \underset{X}{CD)}$
$= X + YZ$
$= CD + AB'B'E$
$= CD + AB'E$

(c) $ACF + AC'F$
$= AF(C+C') = AF$

(d) $\underset{X}{A}(\underset{Y}{C+D'B}) + \underset{X'Z}{A'\cdot 1}$
$= (A+1)(A'+C+D'B)$
$= 1\cdot(A'+C+D'B)$
$= A'+C+D'B$

(e) $(A'B+C+D)(A'B+D)$
令 $X = A'B+D$
$Y = C$
$= (X+Y)X = XX+XY = X+XY = X = A'B+D$

(f) $\underset{X}{(A+BC)} + \underset{Y}{(DE+F)}\underset{X'}{(A+BC)'}$
$= X+Y = (A+BC)+(DE+F)$
$= A+BC+DE+F$ #

1. (a) $757.25_{10} = 1011110101.01_{(2)}$

$$\begin{array}{l} 2\,|\,757 \\ 2\,|\,378 - 1 \\ 2\,|\,189 - 0 \\ 2\,|\,94 - 1 \\ 2\,|\,47 - 0 \\ 2\,|\,23 - 1 \\ 2\,|\,11 - 1 \\ 2\,|\,5 - 1 \\ 2\,|\,2 - 1 \\ 2\,|\,1 - 0 \\ \quad \cancel{0} - 1 \end{array}$$

$$\begin{array}{l} 0.25 \\ \times 2 \\ \hline 0\ \cancel{0}.50 \\ \times 2 \\ \hline 1 - \cancel{1}.00 \end{array}$$

(b) $123.17_{10} = 1111011.0010_{(2)}$

$$\begin{array}{l} 2\,|\,123 \\ 2\,|\,61 - 1 \\ 2\,|\,30 - 1 \\ 2\,|\,15 - 0 \\ 2\,|\,7 - 1 \\ 2\,|\,3 - 1 \\ 2\,|\,1 - 1 \\ \quad 0 - 1 \end{array}$$

$$\begin{array}{l} 0.17 \\ \times 2 \\ \hline 0\ \cancel{0}.34 \\ \times 2 \\ \hline 0\ \cancel{0}.68 \\ \times 2 \\ \hline 1\ \cancel{1}.36 \\ \times 2 \\ \hline 0\ \cancel{0}.72\ \ 2 \end{array}$$

(c) $365.89_{10}$

$= 101101101.11_{(2)}$

$$\begin{array}{l} 2\,|\,365 \\ 2\,|\,182 - 1 \\ 2\,|\,91 - 0 \\ 2\,|\,45 - 1 \\ 2\,|\,22 - 1 \\ 2\,|\,11 - 0 \\ 2\,|\,5 - 1 \\ 2\,|\,2 - 1 \\ 2\,|\,1 - 0 \\ \quad \cancel{0} - 1 \end{array}$$

$$\begin{array}{l} 0.89 \\ \times 2 \\ \hline 1\ \cancel{1}.78 \\ \times 2 \\ \hline 1\ \cancel{1}.56 \end{array}$$

(d) $1063.5_{10} = 10000100111.1_{(2)}$

$$\begin{array}{l} 2\,|\,1063 \\ 2\,|\,531 - 1 \\ 2\,|\,265 - 1 \\ 2\,|\,132 - 1 \\ 2\,|\,66 - 0 \\ 2\,|\,33 - 0 \\ 2\,|\,16 - 1 \\ 2\,|\,8 - 0 \\ 2\,|\,4 - 0 \\ 2\,|\,2 - 0 \\ 2\,|\,1 - 0 \\ \quad \cancel{0} - 1 \end{array}$$

$$\begin{array}{l} 0.5 \\ \times 2 \\ \hline 1\ \cancel{1}.0 \end{array}$$

| A | B | C | D | E | F |
|---|---|---|---|---|---|
| 10 | 11 | 12 | 13 | 14 | 15 |

2.

$3BA.25_{14} = 752.17_{(10)} = 3252.1004_{(6)}$

$$3\times196 + 11\times14 + 10 + 2\times\frac{1}{14} + 5\times\frac{1}{196} = 752.17$$

$$\begin{array}{l} 6\,|\,752 \\ 6\,|\,125 - 2 \\ 6\,|\,20 - 5 \\ 6\,|\,3 - 2 \\ \quad 0 - 3 \end{array}$$

$$\begin{array}{l} 0.17 \\ \times 6 \\ \hline 1\ \cancel{1}.02 \\ \times 6 \\ \hline 0\ \cancel{0}.12 \\ \times 6 \\ \hline 0\ \cancel{0}.72 \\ \times 6 \\ \hline 4\ \cancel{4}.32 \end{array}$$

3. (a)

```
      1111
    × 1010
  --------
     11110
   11110
  --------
  10010110
```

(b)

```
          110110
        × 11101
     -----------
          110110
        11011000
       110110
      110110
     -----------
     11000011110
```

(c)

```
        100100
         10110
     ----------
       1001000
      100100
    1001000
     ----------
    1100011000
```

4. 2's com 6 bits

$-2^{n-1} \sim 2^{n-1}-1$

$= -2^{5} \sim +2^{5}-1$

$= -32 \sim +31$

(a) $21+11$

$21_{10} = 010101_{(2's)}$

$11_{10} = 001011_{(2's)}$

$$\begin{array}{r} 010101 \\ +\ 001011 \\ \hline (1)000000 \end{array}$$

overflow!

(b) $(-14)+(-32) = -56$

$14_{10} = 001110_{(2)}$

$-14_{10} = 110010_{(2's)}$

$32_{10} = 100000_{(2)}$

$-32_{10} = 100000_{(2's)}$

$$\begin{array}{r} 110010 \\ +\ 100000 \\ \hline (1)010010 \end{array}$$

overflow!

(c) $(-25)+18 = -7$

$25_{10} = 011001_{(2)}$

$-25_{10} = 100111_{(2's)}$

$18_{10} = 001010_{(2's)}$

$$\begin{array}{r} 100111 \\ +\ 001010 \\ \hline 110001 \end{array}$$

NOT overflow!

(d) $(-12)+13 = -1$

$12_{10} = 001100_{(2)}$

$-12_{(10)} = 110100_{(2's)}$

$13_{10} = 001101_{(2's)}$

$$\begin{array}{r} 110100 \\ +\ 001101 \\ \hline (1)000001 \end{array}$$

NOT overflow!

(e) $(-11)+(-21) = -32$

$11_{10} = 001011_{(2)}$

$-11_{10} = 110101_{(2's)}$

$21_{10} = 001010_{(2)}$

$-21_{10} = 110110_{(2's)}$

$$\begin{array}{r} 110101 \\ +\ 110110 \\ \hline (1)101011 \end{array}$$

NOT overflow!

6. (a) $X(X'+Y) = XY$

$\Rightarrow XX'+XY = 0+XY = XY$

(b) $X+XY = X$

$\Rightarrow X(1+Y) = X\cdot 1 = X$

(c) $XY+XY' = X(Y+Y') = X\cdot 1 = X$

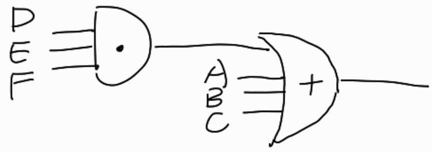
(d) $(A+B)\cdot(A+B') = A\cdot A+AB'+BA+BB'$

$= A+A(B'+B)+0$

$= A+A = A$

8. (a) $(A+B+C+D)(A+B+C+E)(A+B+C+F)$

令 $X=A+B+C$ 令 $D+E=Y$




$= \underbrace{(X+D)(X+E)}(X+F)$

$= (X+XE+XD+DE)(X+F)$

$= X + XF + \underset{XE}{\cancel{XXE}} + XEF + \underset{XD}{\cancel{XXD}} + XDF + XDE + DEF$

$= X + XF(1+E) + XE(1+D) + XD(1+F) + DEF$

$= X + XF \cdot 1 + XE \cdot 1 + XD \cdot 1 + DEF$

$= X + X(DEF) + DEF$

$= X(DEF+1) + DEF$

$= X + DEF = (A+B+C) + DEF$

(b) $WXYZ + VXYZ + UXYZ$

$= (W+V+U)XYZ$